

9 Integrály pohybu

9.1 "Diferenciální" operátorem

$$\hat{A}, \langle \hat{A} \rangle_\psi = \int_{\mathbb{R}} \bar{\psi}(r,t) \hat{A}(x,t) \psi(r,t) dx$$

$$\frac{d}{dt} \langle \hat{A} \rangle_\psi \stackrel{\text{Leibniz}}{=} \int_{\mathbb{R}} \left[\frac{\partial}{\partial t} \bar{\psi} \hat{A} \psi + \bar{\psi} \frac{\partial}{\partial t} \hat{A} \psi + \bar{\psi} \hat{A} \frac{\partial}{\partial t} \psi \right] dx$$

$$\downarrow \text{2 CSR: it } \frac{\partial \psi}{\partial t} = \hat{H} \psi \quad \left| \quad \frac{\partial \bar{\psi}}{\partial t} = \frac{1}{i\hbar} \hat{H} \bar{\psi} \quad \left| \quad \frac{\partial \bar{\psi}}{\partial t} = -\frac{1}{i\hbar} \hat{H} \bar{\psi} \right. \right.$$

$$= \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle_\psi - \frac{1}{i\hbar} \langle \hat{H} \bar{\psi}, \hat{A} \psi \rangle + \frac{1}{i\hbar} \langle \bar{\psi}, \hat{A} \hat{H} \psi \rangle$$

$$= \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle_\psi + \frac{1}{i\hbar} \left[\langle \bar{\psi}, \hat{A} \hat{H} \psi \rangle - \langle \bar{\psi}, \hat{H} \hat{A} \psi \rangle \right]$$

$$= \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle_\psi + \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle_\psi$$

$$\Rightarrow \text{definice: myslíme si, že } \frac{d}{dt} \hat{A} = \frac{\partial \hat{A}}{\partial t} + \frac{1}{i\hbar} [\hat{A}, \hat{H}]$$

$$\frac{\partial \hat{A}}{\partial t} = 0 \Rightarrow \frac{d\hat{A}}{dt} = \frac{1}{i\hbar} [\hat{A}, \hat{H}]$$

$$\frac{d(\hat{A} + \hat{B})}{dt} = \frac{d\hat{A}}{dt} + \frac{d\hat{B}}{dt} \quad \rightarrow \text{se součtem a diferenciálujeme}$$

$$\Rightarrow \hat{x}, \hat{p}: \quad \frac{d\hat{x}}{dt} = \frac{1}{i\hbar} [\hat{x}, \hat{H}]$$

$$\frac{d\hat{p}}{dt} = \frac{1}{i\hbar} [\hat{p}, \hat{H}]$$

$$\hat{H} = \frac{1}{2m} (\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2) + V(x, y, z, t)$$

$$\Rightarrow \frac{d\hat{x}}{dt} = \frac{1}{2m i\hbar} [\hat{x}, \hat{p}_x^2 - \hat{p}_x^2 \hat{x}] =$$

$$= \frac{1}{2m i\hbar} \left[(i\hbar + \hat{p}_x \hat{x}) \hat{p}_x - \hat{p}_x^2 \hat{x} \right]$$

$$= \frac{1}{2m i\hbar} 2i\hbar \hat{p}_x = \frac{\hat{p}_x}{m} \quad \leftrightarrow \quad \frac{d\langle \hat{x} \rangle_t}{dt} = \frac{\langle \hat{p}_x \rangle_t}{m}$$

$$\Rightarrow \frac{d\hat{p}_x}{dt} = \frac{1}{i\hbar} (\hat{p}_x \hat{V} - \hat{V} \hat{p}_x) = \frac{1}{i\hbar} (-i\hbar \frac{\partial V}{\partial x})$$

$$= -\frac{\partial V}{\partial x} =: F_x \quad \leftrightarrow \quad \frac{d\langle \hat{p}_x \rangle_t}{dt} = -\langle \frac{\partial V}{\partial x} \rangle_t$$

$$\frac{d^2 \langle \hat{x} \rangle_t}{dt^2} = \frac{d}{dt} \frac{\langle \hat{p}_x \rangle_t}{m} = \frac{1}{m} \langle \hat{F}_x \rangle_t$$

$$\Rightarrow m \frac{d^2 \langle \hat{x} \rangle_t}{dt^2} = \langle \hat{F}_x \rangle_t$$

9.3 Zákon zachování

$$\hat{A} \text{ je integrálem pohybu} \quad \frac{d\hat{A}}{dt} = 0$$

$$\frac{\partial \hat{A}}{\partial t} = 0 \Rightarrow [\hat{A}, \hat{H}] = 0$$

a) **Volná částice** - je ideál, který není dostatečný

$$\hat{H} = \frac{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}{2m}$$

$$\Rightarrow \hat{p}_x, \hat{p}_y, \hat{p}_z, \hat{H} \text{ jsou IP}$$

$$\rightarrow \text{ZZH a ZZE}$$

$$b) \frac{d\hat{H}}{dt} = \frac{\partial \hat{H}}{\partial t} + \frac{1}{i\hbar} [\hat{H}, \hat{H}] = \frac{\partial \hat{H}}{\partial t}$$

$$\Rightarrow \text{pokud } \hat{H} \text{ nezávisí na čase, pak je ZZE}$$

c) **Centrální pole**

$$V(x, y, z, t) = V(r)$$

$$\hat{H} = -\frac{\hbar^2}{2m} \Delta + V(r) \Rightarrow \text{přechod do sférických souřadnic}$$

$$x = r \sin \theta \cos \varphi \quad r \in [0, +\infty)$$

$$y = r \sin \theta \sin \varphi \quad \theta \in [0, \pi]$$

$$z = r \cos \theta \quad \varphi \in [0, 2\pi)$$

$$dx dy dz = r^2 \sin \theta dr d\theta d\varphi$$

$$\Delta_{r,\theta,\varphi} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \Delta_{\theta,\varphi}$$

$$\Delta_{\theta,\varphi} = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}$$

$$\text{operátor hybnosti: } \hat{L} = \hat{r} \times \hat{p} = (\hat{L}_x, \hat{L}_y, \hat{L}_z)$$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

hledí vyjádřit $\hat{p}$ ve sfér. souř., vyplývá:

$$\hat{L}_x = \hat{y} \hat{p}_z - \hat{z} \hat{p}_y = i\hbar \left(\sin \varphi \frac{\partial}{\partial \theta} + \cot \theta \cos \varphi \frac{\partial}{\partial \varphi} \right)$$

$$\hat{L}_y = \hat{z} \hat{p}_x - \hat{x} \hat{p}_z = -i\hbar \left(\cos \varphi \frac{\partial}{\partial \theta} - \cot \theta \sin \varphi \frac{\partial}{\partial \varphi} \right)$$

$$\hat{L}_z = \hat{x} \hat{p}_y - \hat{y} \hat{p}_x = -i\hbar \frac{\partial}{\partial \varphi}$$

$$\Rightarrow \hat{L}^2 = -\hbar^2 \Delta_{\theta,\varphi}$$

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{\hat{L}^2}{2mr^2} + V(r)$$

zřejmě $\hat{L}^2$ komutuje s $\hat{H}$, jelikož nezávisí na r

$$\Rightarrow [\hat{L}^2, \hat{H}] = [\hat{L}_x, \hat{H}] = [\hat{L}_y, \hat{H}] = [\hat{L}_z, \hat{H}] = 0$$

$\rightarrow$ ZZMH $\Rightarrow$ Keplerovy zákony